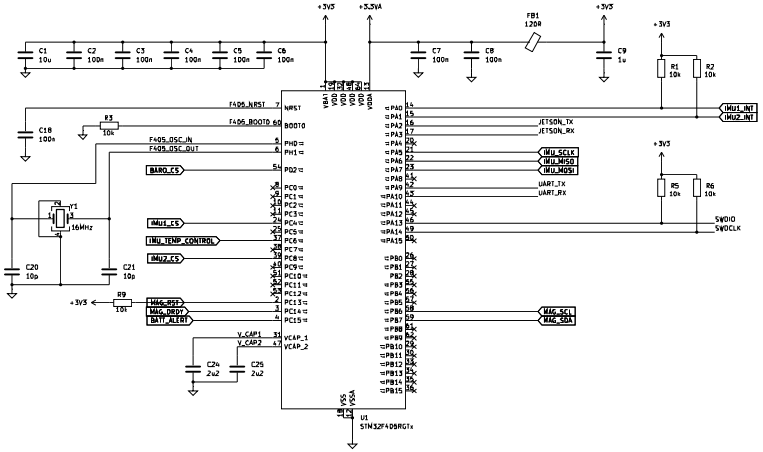
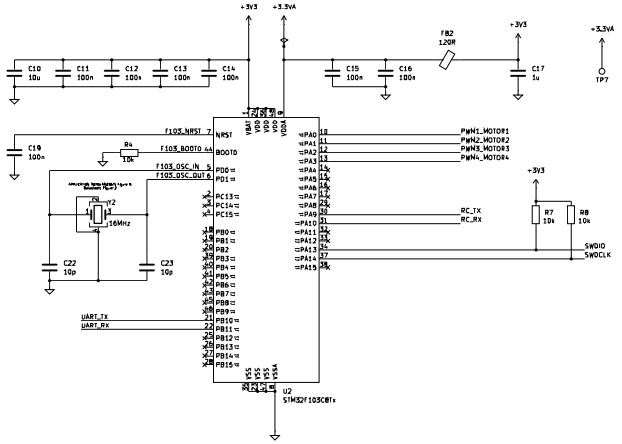
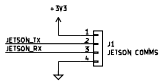
STM32F405 MCU

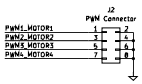


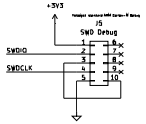
STM32F103 MCU

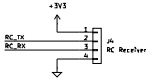


Connectors











IMU

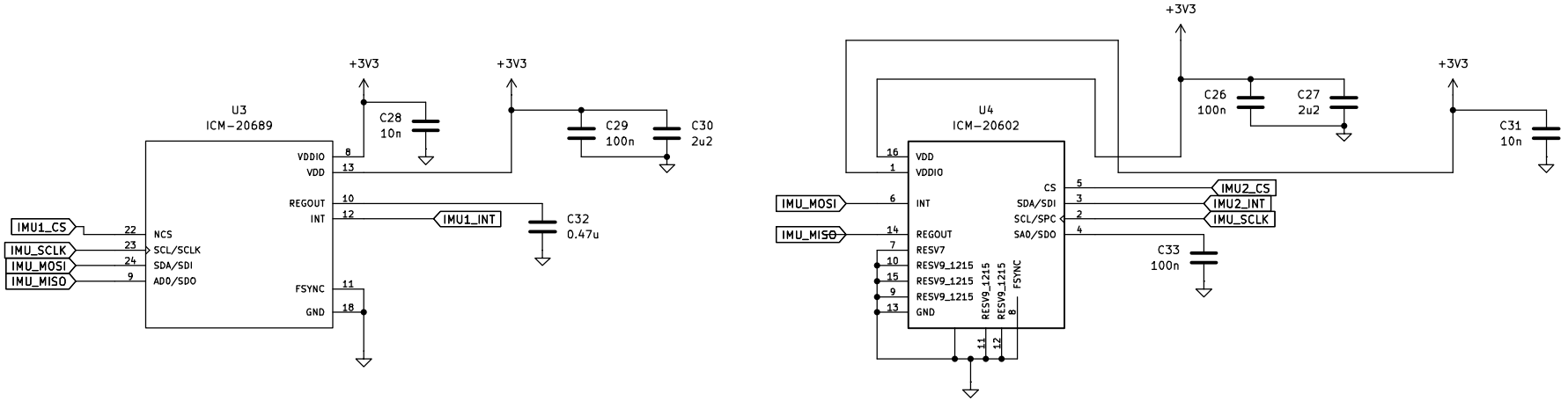
File: IMU_FlightControllerv1.0.kicad_sch

PSM

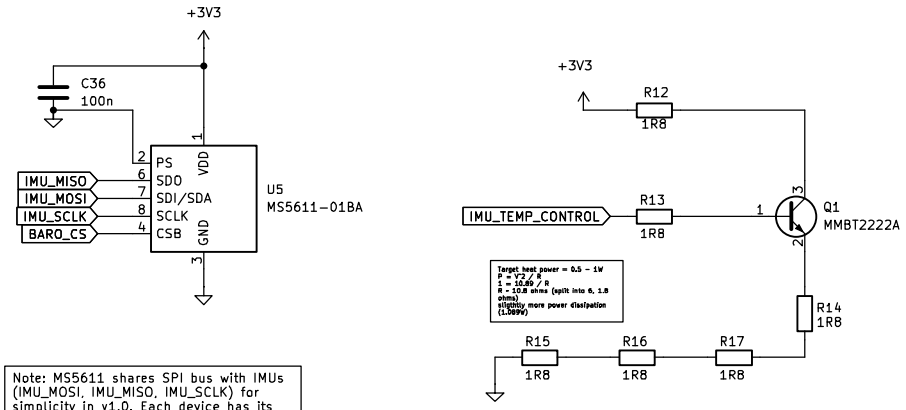
File: PSM_FlightControllerv1.0.kicad_sch

Sheets: /		
File: FlightControllerv1.0.kicad_sch		
Title:		
Size: A2	Date:	Rev:
KiCad E.D.A. 9.0.2		Ids 1/3

IMUs

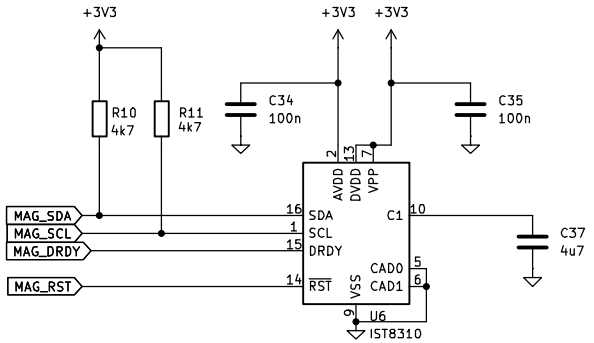


Barometer + IMU Temperature Control



Note: MS5611 shares SPI bus with IMUs (IMU_MOSI, IMU_MISO, IMU_SCLK) for simplicity in v1.0. Each device has its own dedicated chip select (BARO_CS) to prevent bus conflicts. For v1.1 I'll move barometer to a dedicated SPI bus (BARO_MOSI, BARO_MISO, BARO_SCLK) to isolate pressure readings from IMU switching noise

Magnometer

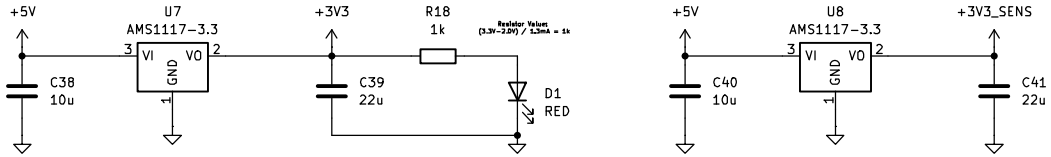


Sheet: /IMU/
File: IMU_FlightController.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.2		Id: 2/3

LDOs

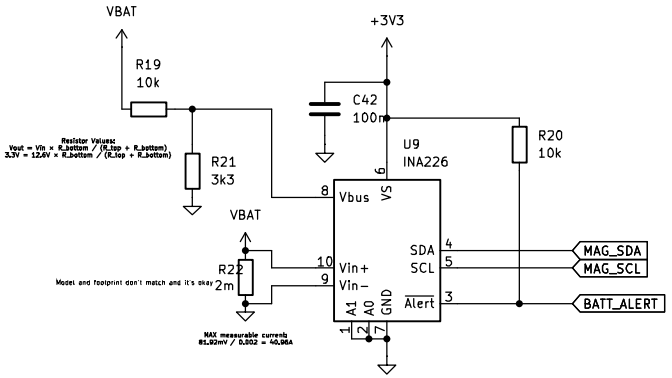


LDO 1:
Powers the STM32F405, STM32F103, memory chips, connectors,
basically all the digital logic on the board.

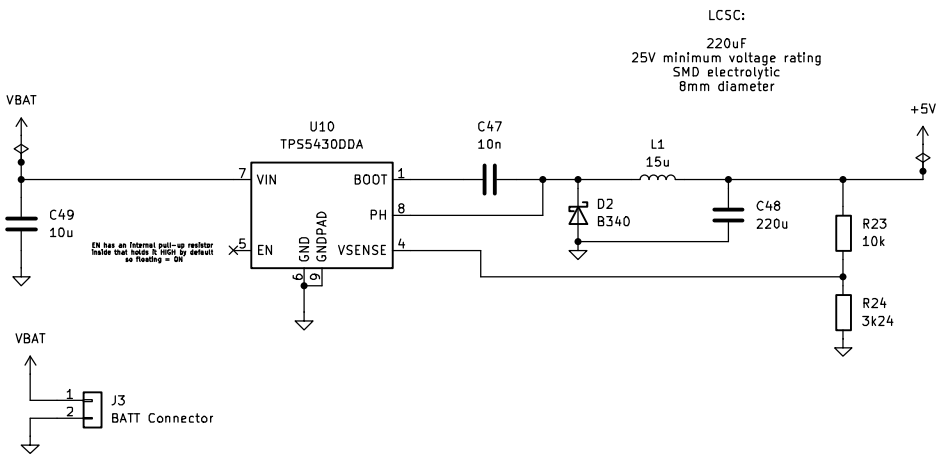
LDO 2:
Powers only the IMUs, barometer, and magnetometer
completely isolated from the logic rail.

Notes:
The output cap is what maintains stability of the regulated output voltage and filters noise on the 3.3V rail.
LDOs are actually more sensitive to output capacitance than input capacitance — a larger output cap gives better transient response when the load suddenly changes, like when the STM32 wakes up a peripheral.
The AMS1117 specifically requires a minimum output capacitance to stay stable. The input cap just needs to be large enough to prevent the input voltage from drooping when the LDO draws current — 10uF is enough for that job.
(These values were recommended by the datasheet)

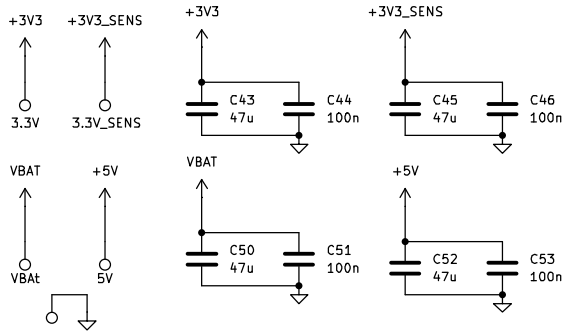
V/C Monitor



Buck Regulator



Decop Caps + Test Points



Sheet: /PSM/
File: PSM_FlightController_sch.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 3/3